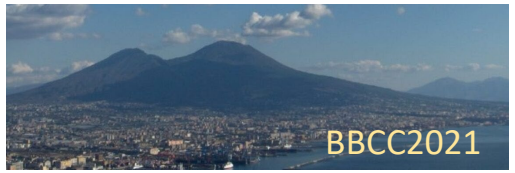
EpiStatProfiler: a new R package for the qualitative analysis of DNA methylation



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Motivation

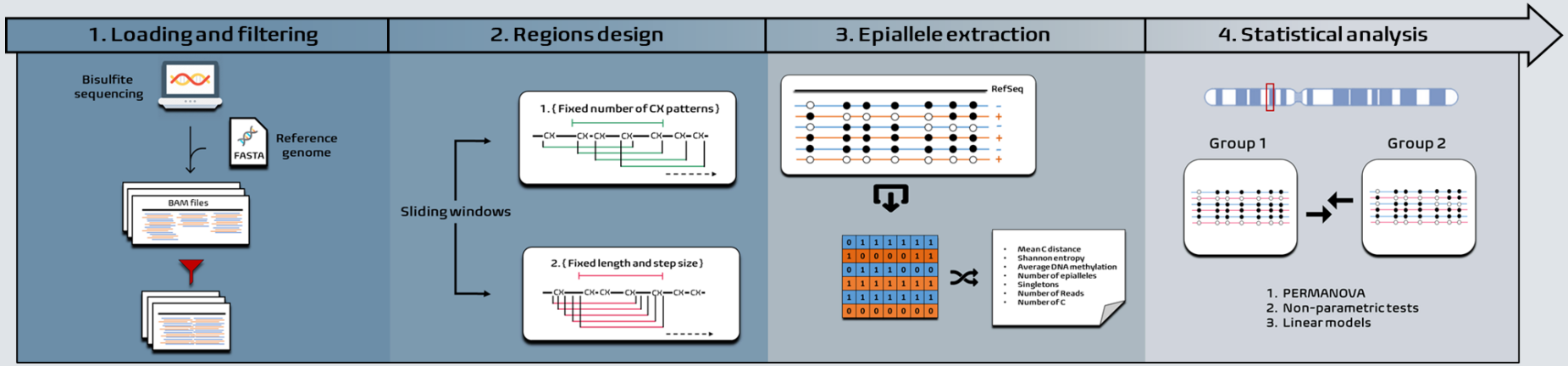
DNA methylation is an epigenetic modification, primarily occurring at CpG sites, involved in major biological mechanisms. Typically, association studies are aimed to identify genomic regions whose average DNA methylation differs among distinct conditions. However, the qualitative analysis of DNA methylation, analyzing all the observed combinations of CpGs methylation states in one locus (***epialleles***), can provide additional insights about the cell-to-cell heterogeneity and the cell clonality within a sample. Various bioinformatic tools have been developed to extract the epiallele information from bisulfite sequencing data, but few are designed for genome-wide experiments and none of them provide dedicated statistical tests needed to compare epiallele composition among different groups.

Methods

Epialleles are extracted from bisulfite sequencing data in different steps. Filtering steps have been developed to select regions of interest. Two distinct methods have been designed to build target regions. Different statistical tests have been used to construct functions aimed to identify the most significant regions among distinct groups.

Results

We developed EpiStatProfiler, a new R-package providing a library of functions that can be used to extract and summarize epialleles from any type of bisulfite sequencing data and to perform downstream statistical comparisons among different groups. Our tool is also designed to analyze strand-specific and non-CG DNA methylation.



EpiStatProfiler workflow. The figure shows the main steps of the novel presented tool. EpiStatProfiler is intended to enable a customized selection of target regions, according to a set of user-defined parameters (minimum required coverage, number of CX sites to analyse, minimum window size) through each of the steps described above.